

Computational Fluid Dynamics for Aerospace Applications

Lecture 2 Governing Equations 1

Governing Equations

- Mass
- Momentum
- Types of Terms

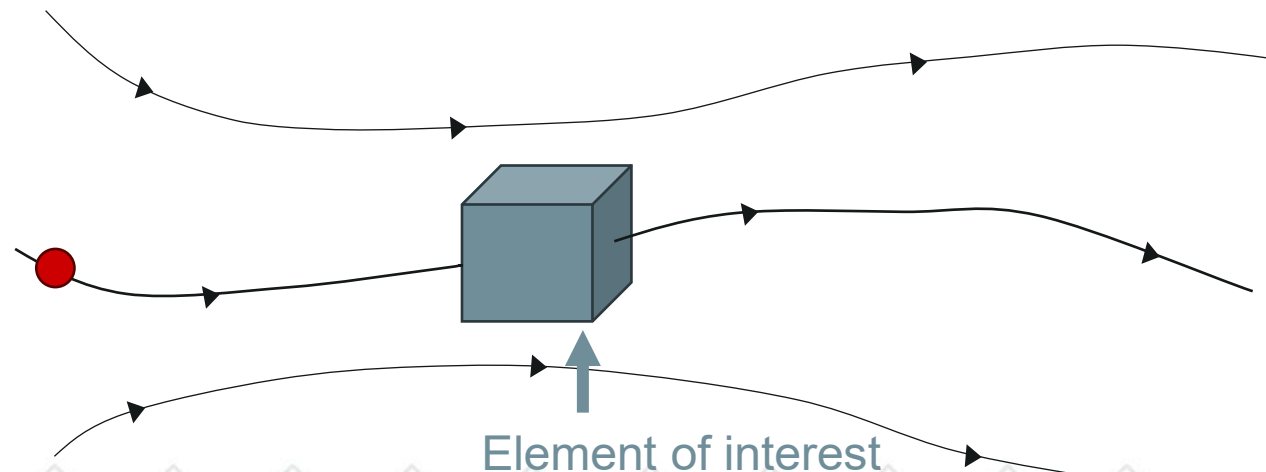
Governing Equations

- Computational Fluid Dynamics is Fluid Dynamics
 - Must understand what equations to solve and what they mean
- For this class, we will focus on compressible fluids
 - Primarily relevant for aerospace applications
 - Will briefly touch on incompressible (easier equations, harder to solve)
- Many different ways to derive and express equations
 - Lagrangian, Eulerian, Conservative, Nonconservative, etc.
 - We are focusing on one form that gets us to a useable result quickly

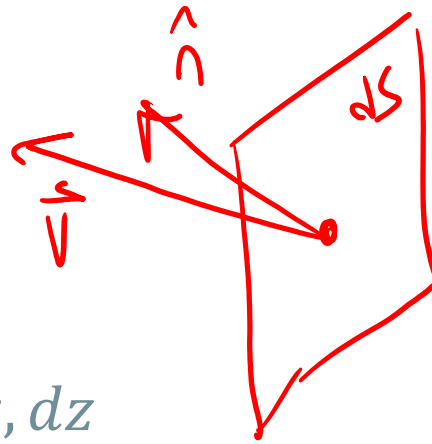
Mass

Net mass flow rate out of volume across surface = Time rate of decrease of mass inside volume

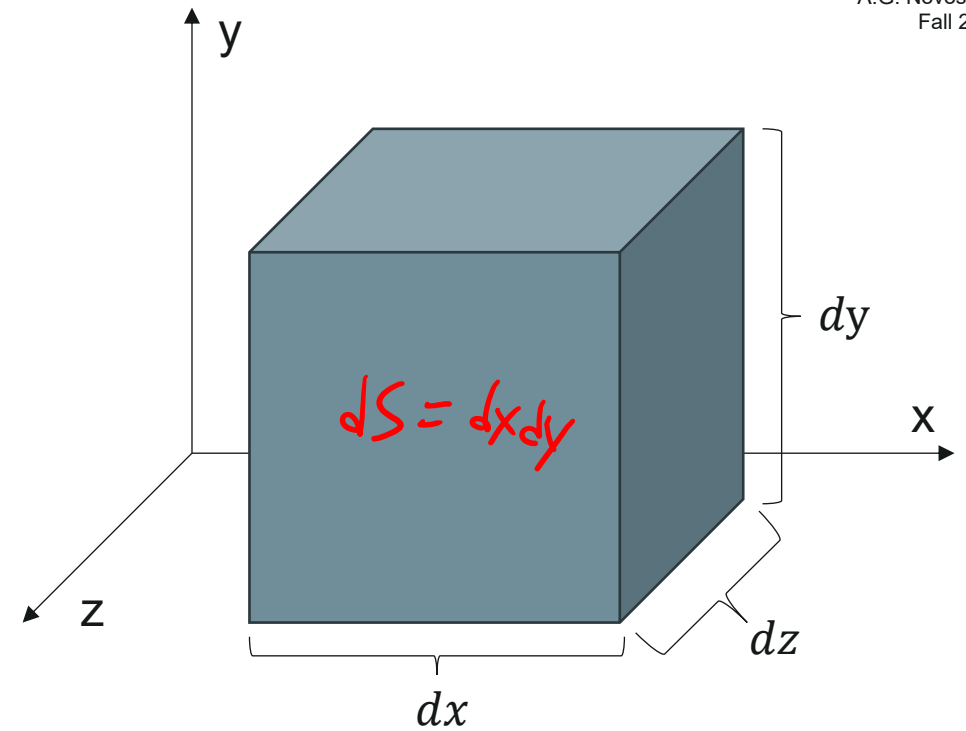
- Mass equation is derived from first principles
 - Mass is conserved
- Analyze an infinitesimally small element fixed in space
- Cartesian coordinates
- Velocity and density are functions of space and time
 - $\rho = \rho(x, y, z, t)$
 - $\mathbf{V} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$
 - $u = u(x, y, z, t)$
 - $v = v(x, y, z, t)$
 - $w = w(x, y, z, t)$



Mass



- Our element has
 - Side lengths dx, dy, dz
 - Volume dV
- How much mass crosses an arbitrary surface per time?



$\vec{V} \cdot \hat{n} [l/t] \rightarrow$ Velocity normal to surface

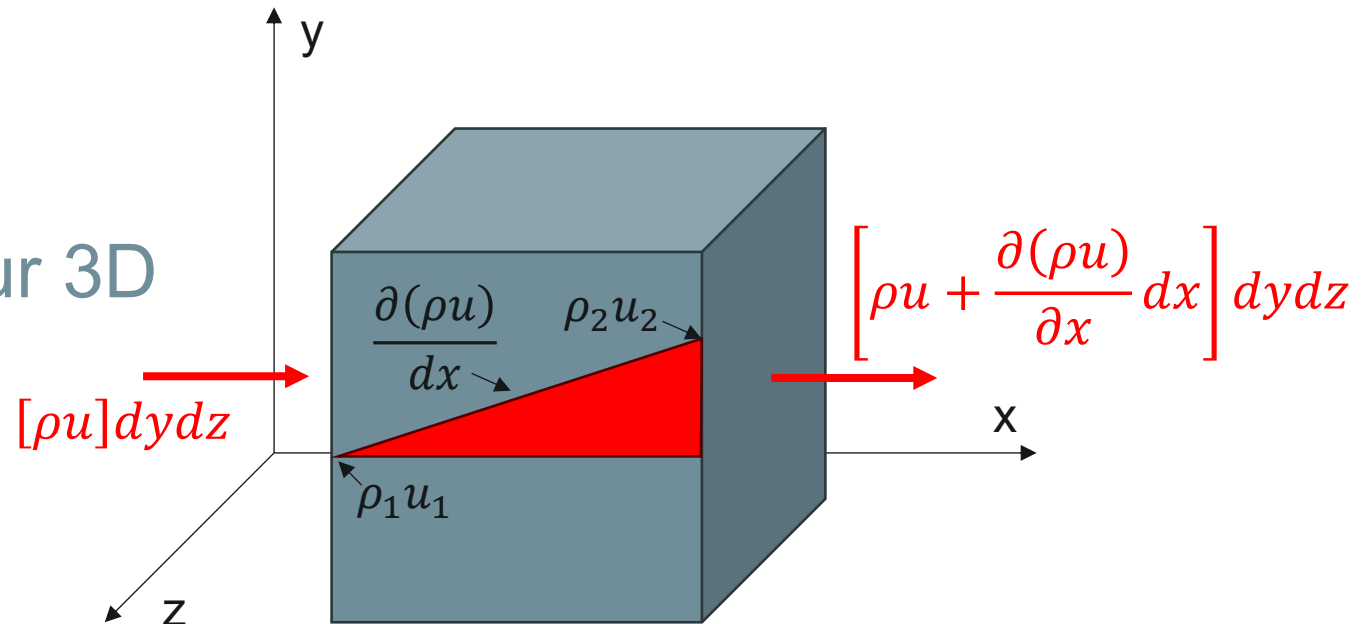
$\vec{V} \cdot \hat{n} dS [l^3/t] \rightarrow$ volume per time

$\rho \vec{V} \cdot \hat{n} dS [m/l^3] [l^3/t] = [m/t] \rightarrow$ mass per time

$$\rho \vec{V} \cdot \hat{n} dS$$

Mass

- How much mass crosses our 3D element in the x-direction?



Left face:

$$dS = dz dy$$

$$\rho \vec{V} \cdot \hat{n} = \rho u$$

Right face

$$dS = dz dy$$

$$\rho \vec{V} \cdot \hat{n} = \left[\rho u + \frac{\partial(\rho u)}{\partial x} dx \right]$$

Mass crossing in x-direction = net mass out left + net mass out right

$$= - \rho \vec{V} \cdot \hat{n} dS \Big|_{\text{left}}^{\text{in}} + \rho \vec{V} \cdot \hat{n} dS \Big|_{\text{right}}^{\text{out}}$$

$$= - \rho u dz dy + \left[\rho u + \frac{\partial(\rho u)}{\partial x} dx \right] dz dy$$

$$\frac{\partial(\rho u)}{\partial x} dx dy dz$$

Mass

- Net mass flow rate out

$$\underline{x}: \frac{\partial(\rho u)}{\partial x} dx dy dz$$

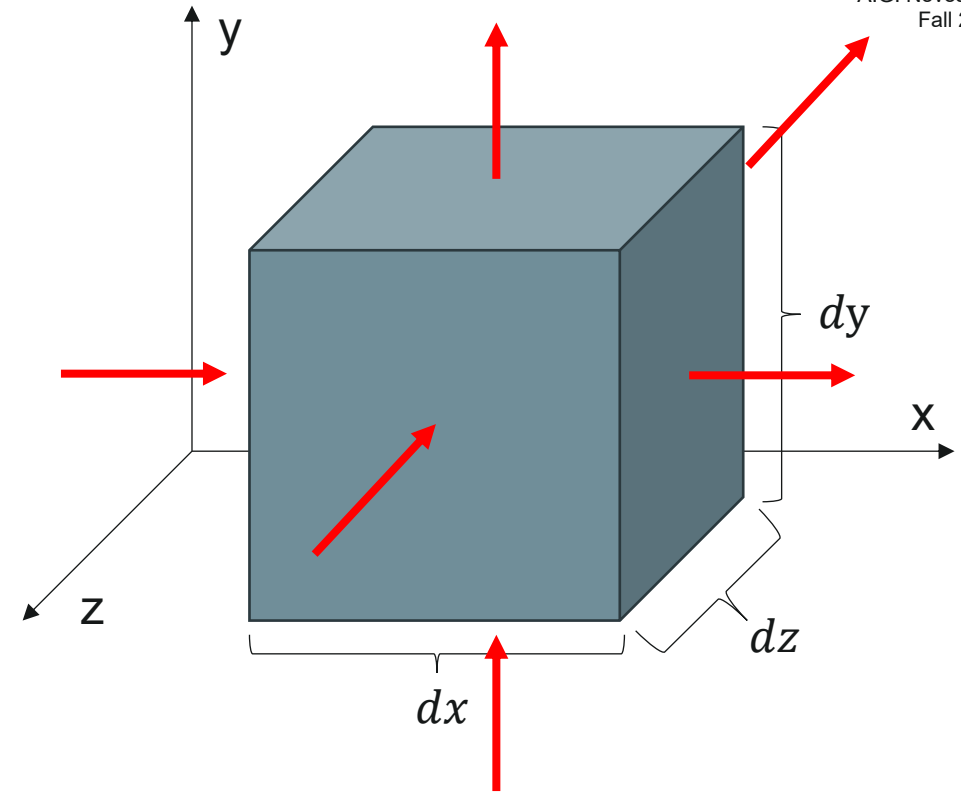
$$\underline{y}: \frac{\partial(\rho v)}{\partial y} dx dy dz$$

$$\underline{z}: \frac{\partial(\rho w)}{\partial z} dx dy dz$$

Net mass
flow out =
 $\text{Sum}(x, y, z)$



$$\left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] dx dy dz$$



Mass

- Continuity equation (mass conservation)

Net mass flow rate out of volume across surface = Time rate of decrease of mass inside volume

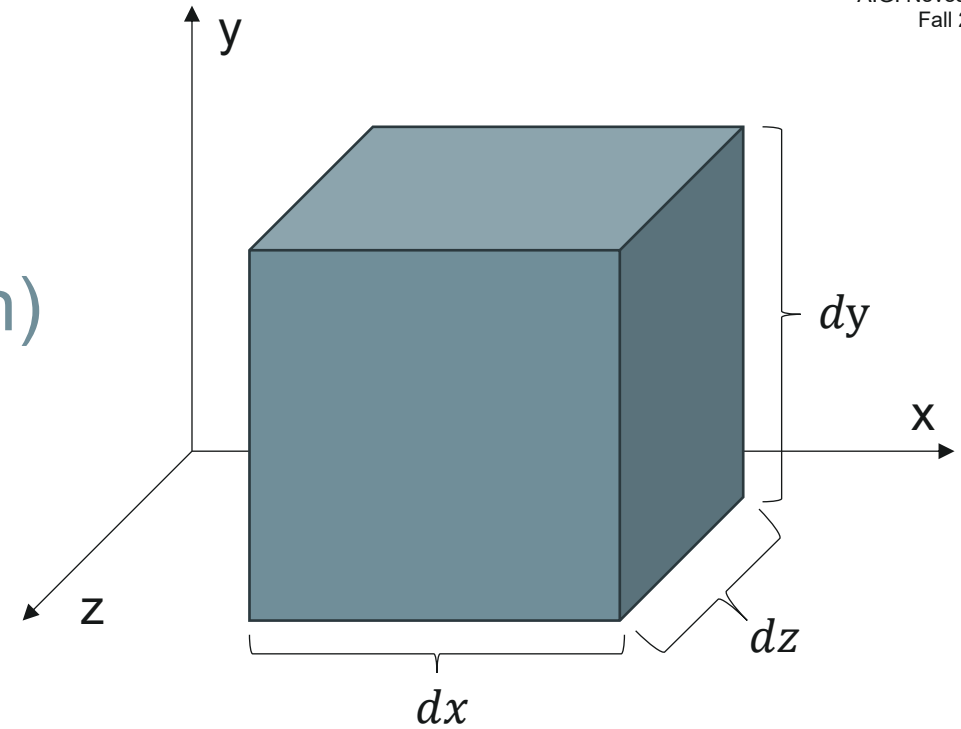
$$\text{mass inside element} = \rho dx dy dz$$

Net rate of change of mass:

$$\frac{\partial}{\partial t} (\rho dx dy dz) = \frac{\partial \rho}{\partial t} dx dy dz$$

mass conservation:

$$\left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] dx dy dz = - \frac{\partial \rho}{\partial t} dx dy dz$$



$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

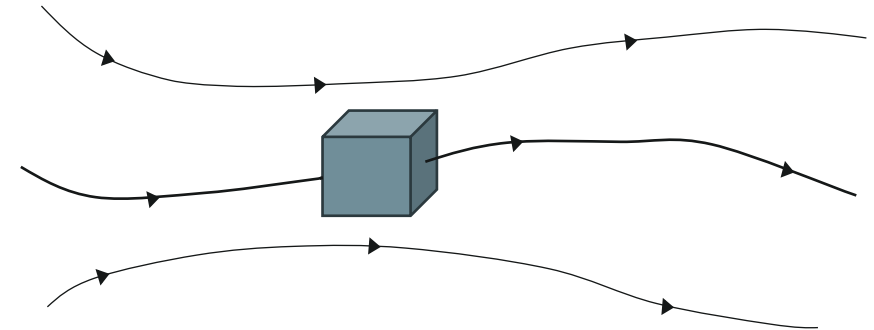
Mass

- Continuity equation (mass conservation)

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

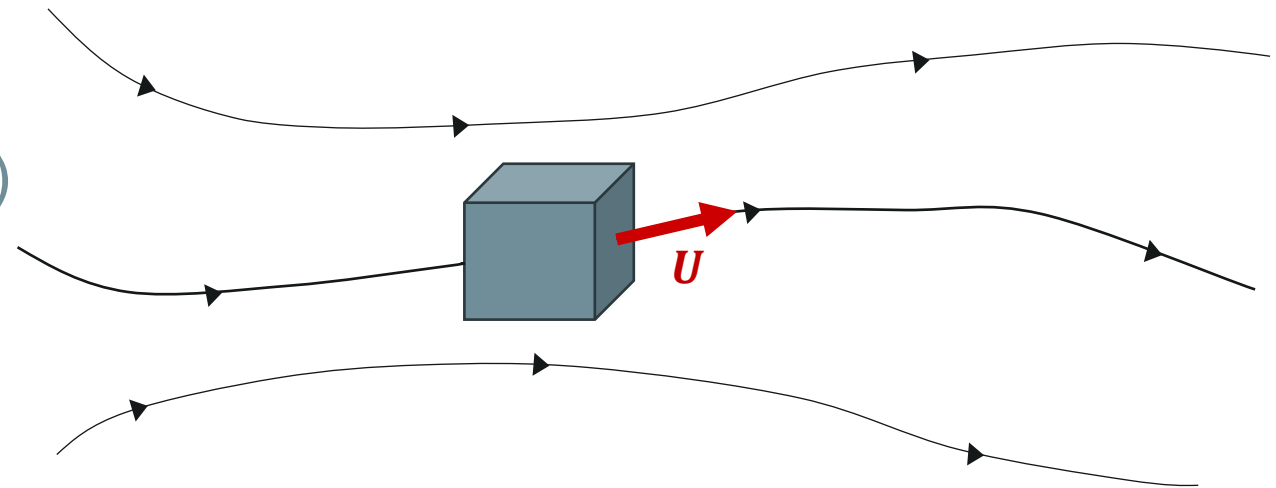
$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

from an
element
fixed in
space



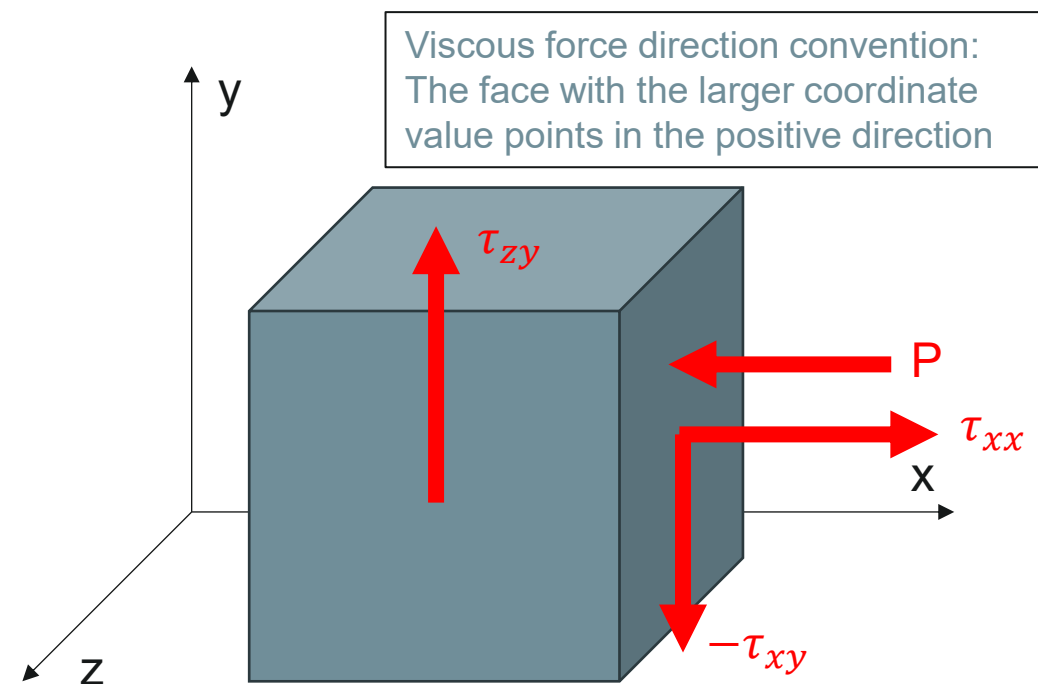
Momentum

- Momentum equations are derived from first principles
 - $F = ma$
 - Analyze an infinitesimally small element moving with the flow
- Forces applied to flow
 - Body Forces (act on volume)
 - Gravity, electric, magnetic, etc.
 - Surface Forces (act on surface)
 - Pressure
 - Viscous
 - Normal
 - Shear



Momentum

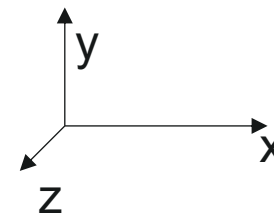
- Focus on x-direction first
 - $F_x = ma_x$
- Body forces in x-direction
 - f_x is body force per unit mass in x
 - Total body force in x is $f_x \rho dx dy dz$
- Pressure forces in x-direction
 - P is the pressure force per unit area
 - Pressure always acts normal to a surface and inwards
- Viscous forces in x-direction *per area*
 - τ_{ij} is a viscous force on the i surface in the j direction
 - τ_{xy} is a shear stress on the x face pointing in the y-direction
 - τ_{xx} is a normal force on the x face pointing in the x-direction



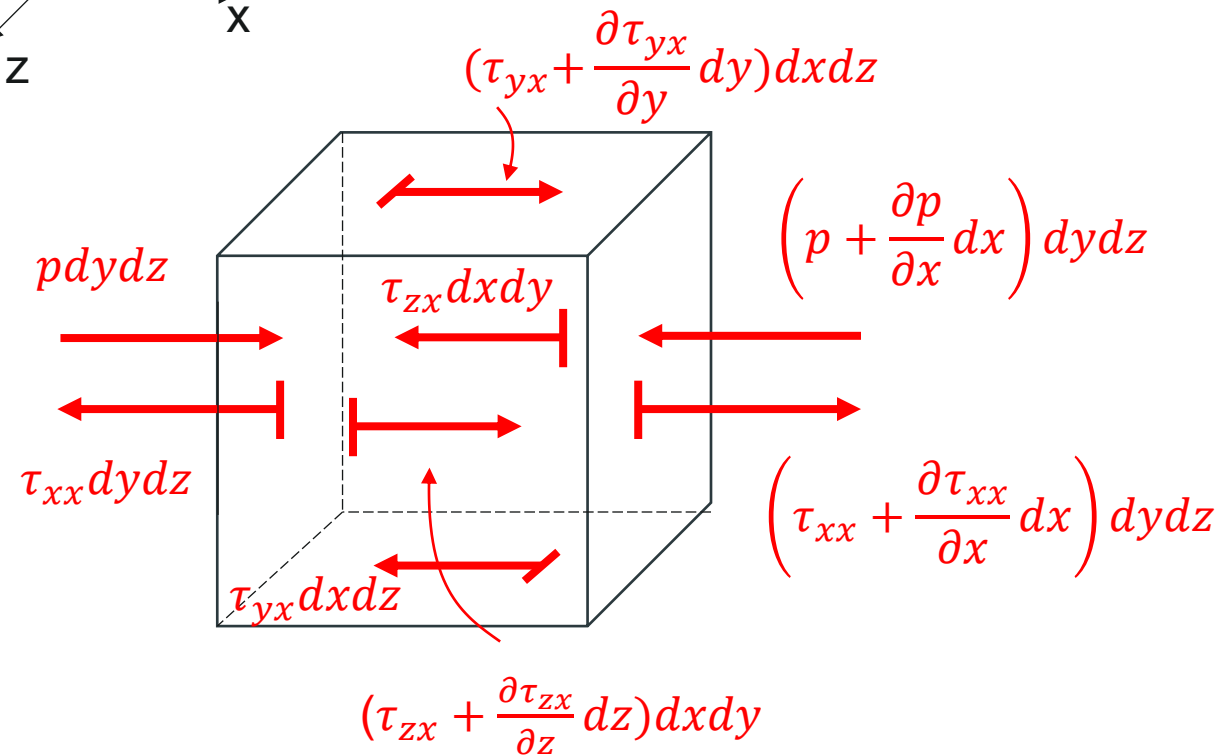
Momentum

• X-direction derivation

$$\begin{aligned}
 F_x = & \left[\cancel{p} - \left(\cancel{p} + \frac{\partial p}{\partial x} dx \right) \right] dy dz \\
 & + \left[\left(\cancel{\tau_{xx}} + \frac{\partial \tau_{xx}}{\partial x} dx \right) - \cancel{\tau_{xx}} \right] dy dz \\
 & + \left[\left(\cancel{\tau_{zx}} + \frac{\partial \tau_{zx}}{\partial z} dz \right) - \cancel{\tau_{zx}} \right] dx dy \\
 & + \left[\left(\cancel{\tau_{yx}} + \frac{\partial \tau_{yx}}{\partial y} dy \right) - \cancel{\tau_{yx}} \right] dx dz \\
 & + \rho f_x dx dy dz
 \end{aligned}$$



Viscous force direction convention:
The face with the larger coordinate value points in the positive direction

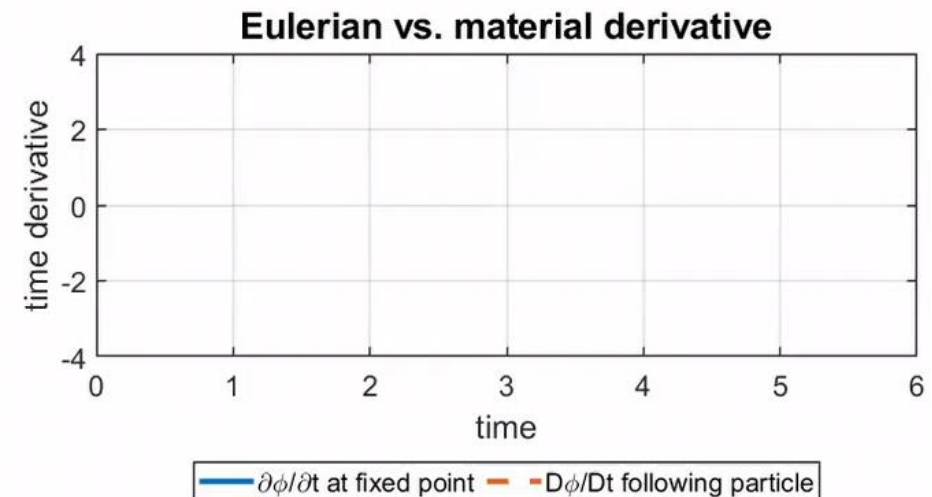
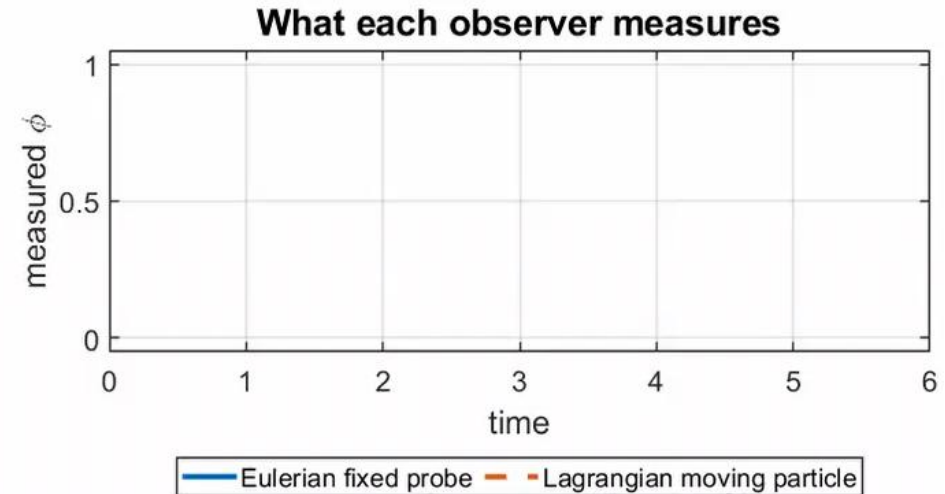
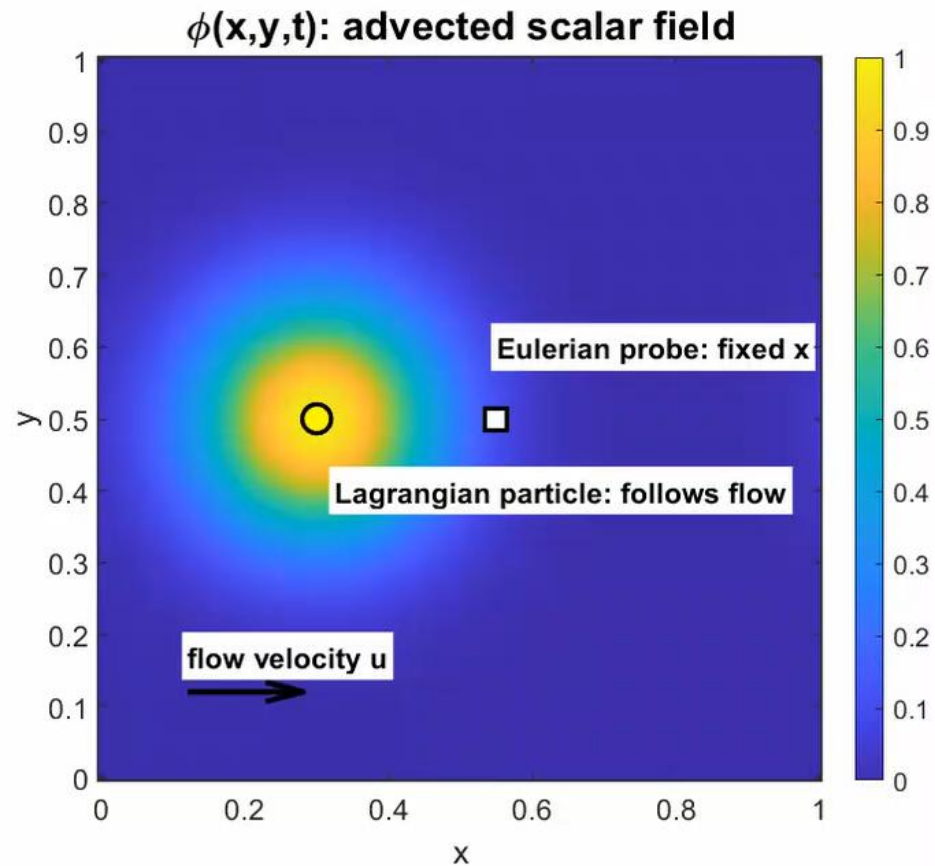


$$F_x = \left(\left[-\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right] + \rho f_x \right) dx dy dz$$

Momentum

- A quick note: we will encounter two different types of derivatives working with the Navier-Stokes equations
- Material derivative (D/Dt) describes *Lagrangian* motion, i.e., when we follow a moving particle. How does what the particle experiences change over time?
- Regular derivative (d/dt) describes *Eulerian* motion, i.e., when we focus on a fixed point in space. How does the quantity at that space change over time?

Momentum



Momentum

- The two derivatives can be related in fluid mechanics:

$$\underbrace{\frac{D\phi}{Dt}}_{\text{Change experienced by moving particle}} = \underbrace{\frac{\partial \phi}{\partial t}}_{\text{Change experienced by stationary particle}} + \underbrace{\mathbf{V} \cdot \nabla \phi}_{\text{Change due to motion of the particle/flow}}$$

- Back to X-direction derivation

- $F_x = ma_x$

$$m = \rho \, dx \, dy \, dz$$

$$a_x = \frac{Du}{Dt}$$

$$F_x = m a_x$$

$$\left[\left(-\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) + \rho f_x \right] dx \, dy \, dz$$

$$= \rho \, dx \, dy \, dz \, \frac{Du}{Dt}$$

$$\frac{d(\rho u)}{dt} = \rho \frac{du}{dt} + u \frac{d\rho}{dt} \quad \bigg| \quad \nabla \cdot (\rho \vec{V} u) = u \nabla \cdot \rho \vec{V} + \rho \vec{V} \cdot \nabla u$$

Momentum

• X-direction derivation

$$\textcircled{1} \quad \rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x$$

$$\textcircled{2} \quad \rho \frac{Du}{Dt} = \frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \vec{V})$$

$$\leftarrow \rho \frac{Du}{Dt} = \frac{d(\rho u)}{dt} + (\vec{V} \cdot \nabla \rho \vec{V} u) - u \left[\frac{d\rho}{dt} + \nabla \cdot \rho \vec{V} \right] \stackrel{\text{continuity}}{=} 0$$

$$\textcircled{1} + \textcircled{2} \quad \frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \vec{V}) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x$$

Removing vector notation ↓

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} + \frac{\partial(\rho u v)}{\partial y} + \frac{\partial(\rho u w)}{\partial z} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x$$

From last slide:

$$\frac{Du}{Dt} = \frac{du}{dt} + \vec{V} \cdot \nabla u$$

$$\rho \frac{Du}{Dt} = \boxed{\rho \frac{du}{dt}} + \boxed{\rho \vec{V} \cdot \nabla u}$$

$$= \boxed{\frac{d(\rho u)}{dt} - u \frac{d\rho}{dt}} + \boxed{(\nabla \cdot (\rho \vec{V} u) - u \nabla \cdot \rho \vec{V})}$$

Momentum

- The longest/most confusing part of the derivation is relating viscous stresses to velocity gradients – we will skip this
- Assume Newtonian + Stokes' hypothesis

$$\bullet \tau_{xx} = 2\mu \frac{\partial u}{\partial x} - \frac{2}{3}\mu(\nabla \cdot \mathbf{V})$$

$$\bullet \tau_{xy} = \tau_{yx} = \mu \left[\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right]$$

$$\bullet \tau_{xz} = \tau_{zx} = \mu \left[\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right]$$

Symmetry

Useful for other equations
(i.e., y and z directions)

$$\tau_{yy} = 2\mu \frac{\partial v}{\partial y} - \frac{2}{3}\mu(\nabla \cdot \mathbf{V})$$

$$\tau_{zz} = 2\mu \frac{\partial w}{\partial z} - \frac{2}{3}\mu(\nabla \cdot \mathbf{V})$$

$$\tau_{yz} = \tau_{zy} = \mu \left[\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right]$$

$$\frac{\partial \phi}{\partial t} + \frac{\partial \phi u}{\partial x} = \frac{\partial}{\partial x} \left(\mu \frac{\partial \phi}{\partial x} \right) + S$$

Momentum

The Navier-Stokes Equations

X momentum

$$\underbrace{\frac{\partial(\rho u)}{\partial t}}_{\frac{\partial \phi}{\partial t}} + \underbrace{\frac{\partial(\rho u u)}{\partial x}}_{\frac{\partial \phi u}{\partial x}} + \frac{\partial(\rho u v)}{\partial y} + \frac{\partial(\rho u w)}{\partial z} = -\frac{\partial p}{\partial x} + \underbrace{\frac{\partial}{\partial x} \left(2\mu \frac{\partial u}{\partial x} - \frac{2}{3} \mu (\nabla \cdot \mathbf{V}) \right)}_{\substack{\uparrow \\ S, \text{ kind of}}} + \underbrace{\frac{\partial}{\partial y} \left(\mu \left[\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right] \right)}_{\frac{\partial}{\partial x} \left(\mu \frac{\partial \phi}{\partial x} \right)} + \frac{\partial}{\partial z} \left(\mu \left[\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right] \right) + \underbrace{\rho f_x}_S$$

$\frac{\partial \phi}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$

Y momentum

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho v u)}{\partial x} + \frac{\partial(\rho v v)}{\partial y} + \frac{\partial(\rho v w)}{\partial z} = -\frac{\partial p}{\partial y} + \frac{\partial}{\partial x} \left(\mu \left[\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right] \right) + \frac{\partial}{\partial y} \left(2\mu \frac{\partial v}{\partial y} - \frac{2}{3} \mu (\nabla \cdot \mathbf{V}) \right) + \frac{\partial}{\partial z} \left(\mu \left[\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right] \right) + \rho f_y$$

Z momentum

$$\frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho w u)}{\partial x} + \frac{\partial(\rho w v)}{\partial y} + \frac{\partial(\rho w w)}{\partial z} = -\frac{\partial p}{\partial z} + \frac{\partial}{\partial x} \left(\mu \left[\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right] \right) + \frac{\partial}{\partial y} \left(\mu \left[\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right] \right) + \frac{\partial}{\partial z} \left(2\mu \frac{\partial w}{\partial z} - \frac{2}{3} \mu (\nabla \cdot \mathbf{V}) \right) + \rho f_z$$

Types of Terms

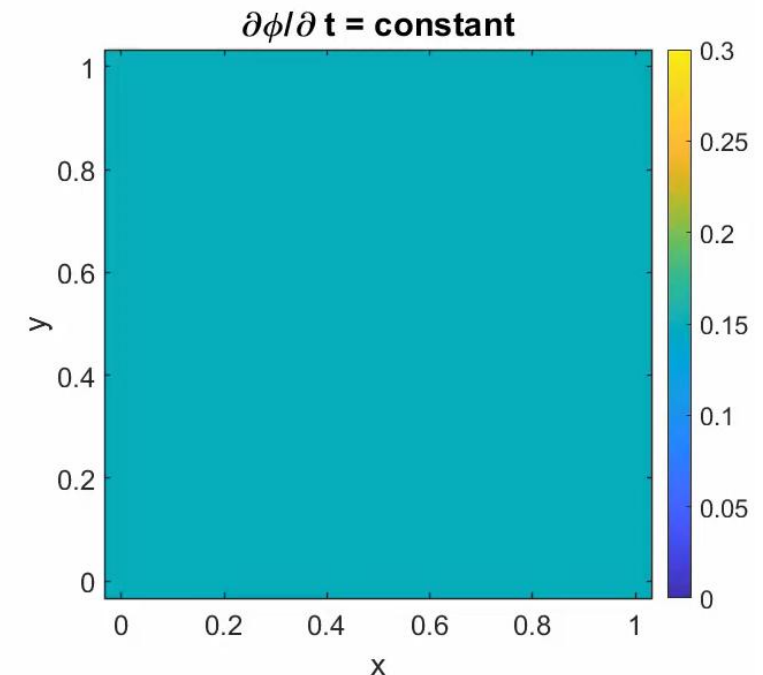
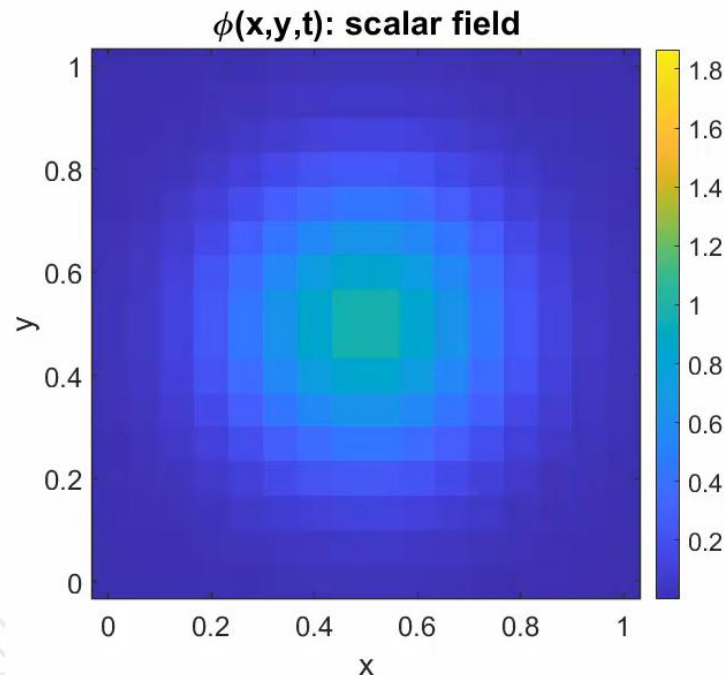
Consider an equation of the form:

$$\frac{\partial(\phi)}{\partial t} + \frac{\partial(\phi u)}{\partial x} = \frac{\partial}{\partial x} \left(\Gamma \frac{\partial \phi}{\partial x} \right) + S_\phi$$

- Unsteady Term (e.g., $\frac{\partial \phi}{\partial t} = \frac{\partial u}{\partial t}$)
 - A local term defining how the quantity changes at one point
 - This is the term we are “tracking” in an unsteady flow – i.e., we want to see how ϕ evolves over time at every point

Unsteady term = constant

$$\frac{\partial \phi}{\partial t} = c$$



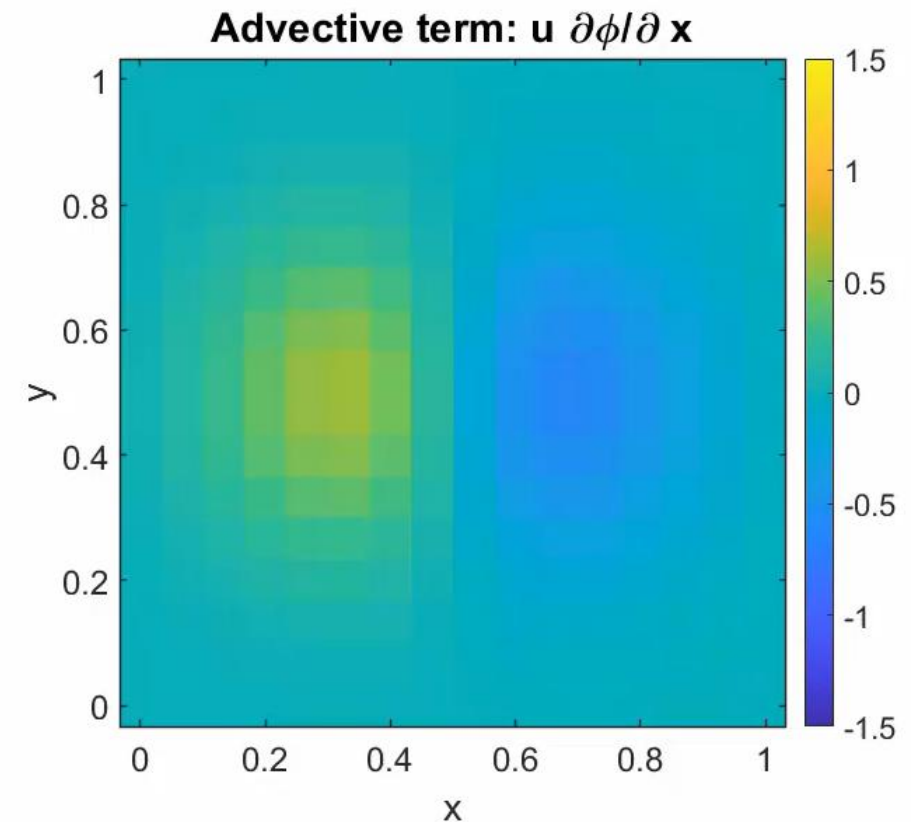
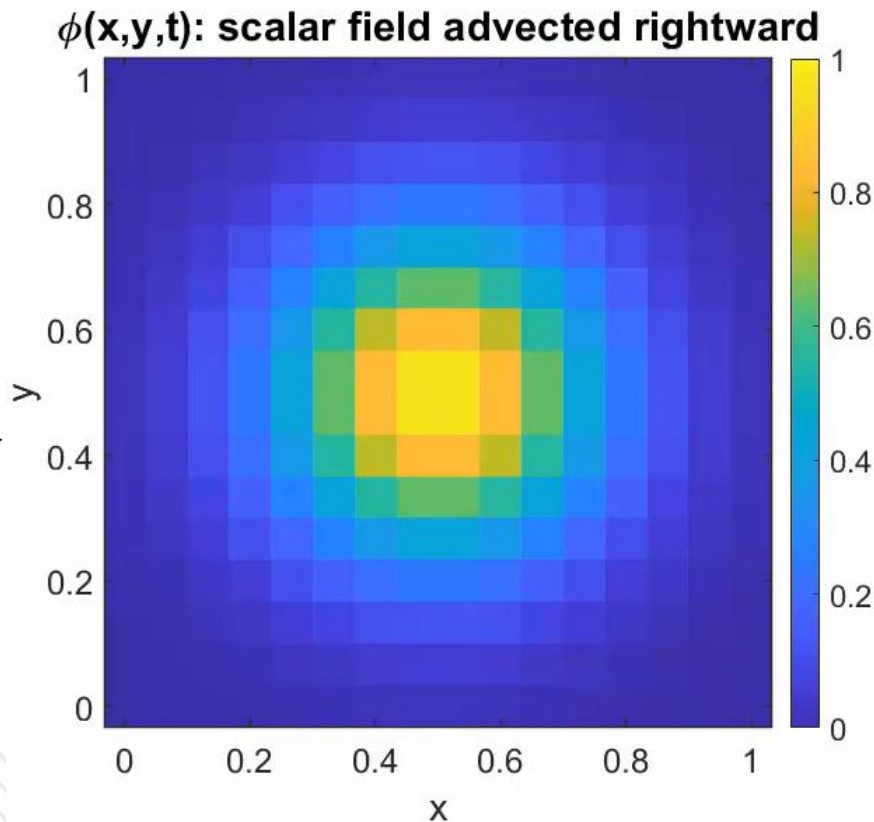
Types of Terms

Consider an equation of the form:

$$\frac{\partial(\phi)}{\partial t} + \frac{\partial(\phi u)}{\partial x} = \frac{\partial}{\partial x} \left(\Gamma \frac{\partial \phi}{\partial x} \right) + S_\phi$$

- Advective Term (or convective, e.g., $\frac{\partial(\phi u)}{\partial x} = \frac{\partial \rho u u}{\partial x}$)
 - A nonlocal term defining bulk motion of a quantity due to flow velocity

Unsteady term =
Advective term



Types of Terms

Consider an equation of the form:

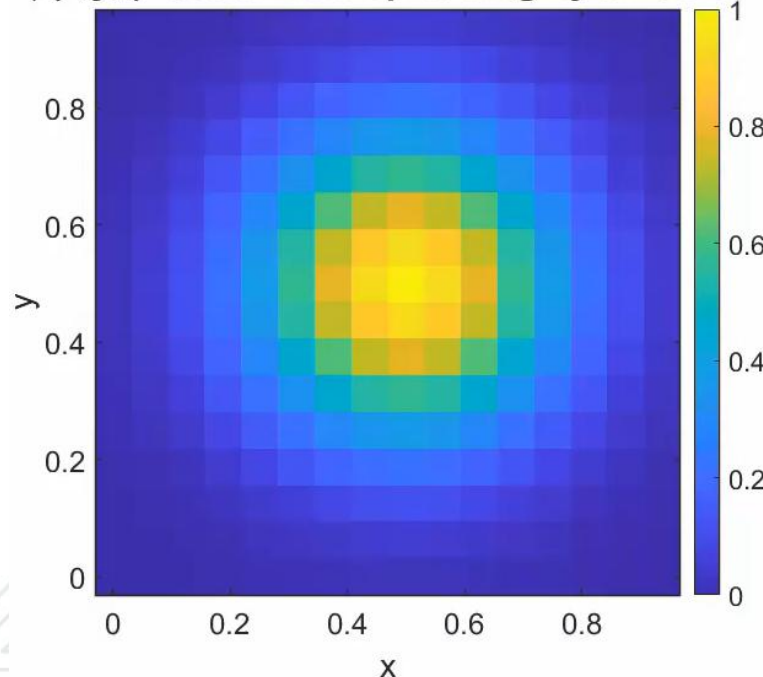
$$\boxed{\frac{\partial(\phi)}{\partial t}} + \frac{\partial(\phi u)}{\partial x} = \boxed{\frac{\partial}{\partial x} \left(\Gamma \frac{\partial \phi}{\partial x} \right)} + S_\phi$$

- Diffusive Term (e.g., $\frac{\partial}{\partial x} \left(\Gamma \frac{\partial \phi}{\partial x} \right) = \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right)$)
 - A nonlocal term defining the movement of a quantity due to gradients in that quantity
 - Γ is a transport coefficient (e.g., μ, D, λ) that describes how fast ϕ diffuses

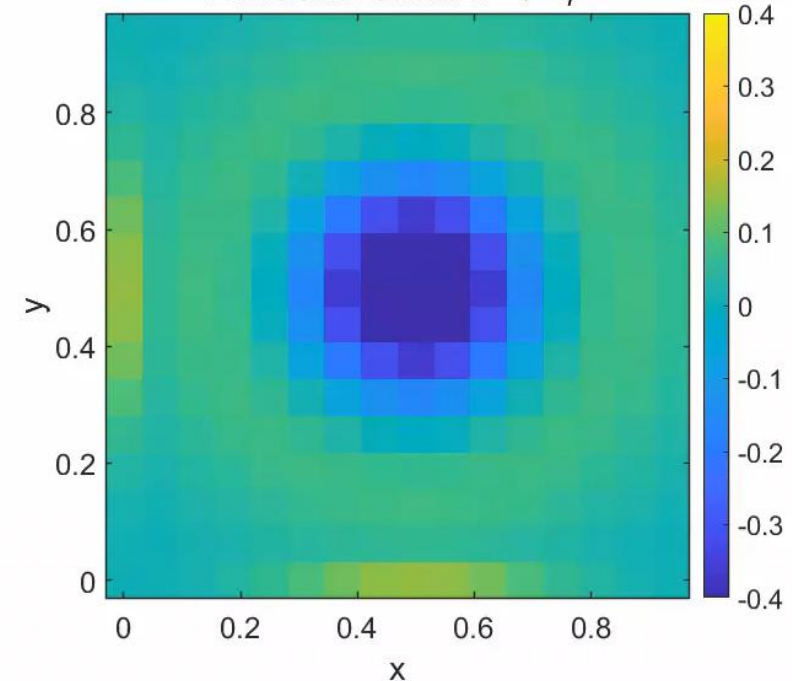
Unsteady term =
Diffusive term



$\phi(x,y,t)$: scalar field spreading by diffusion



Diffusive term: $\Gamma \nabla^2 \phi$



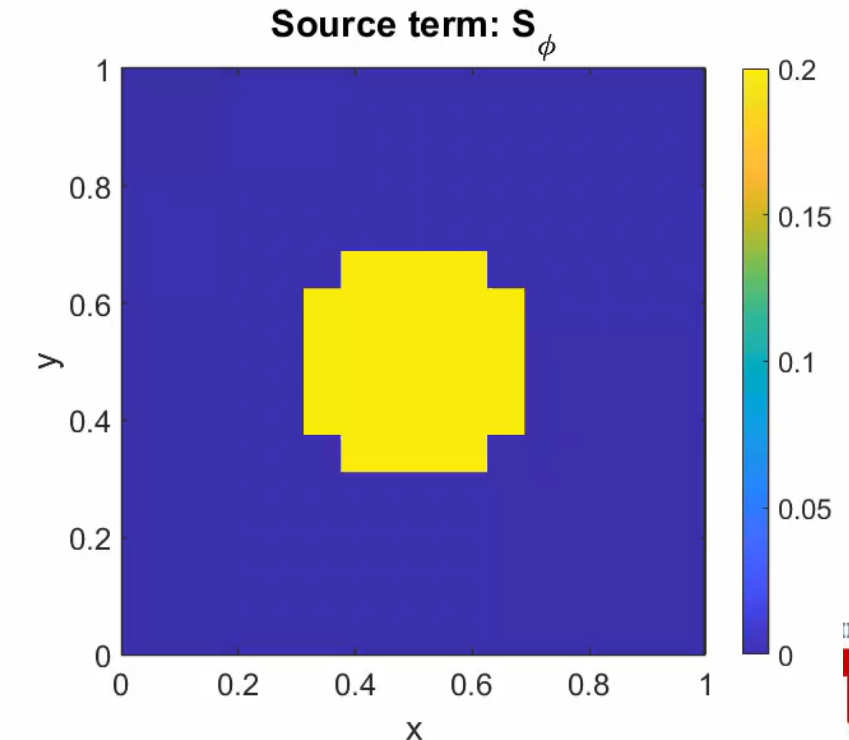
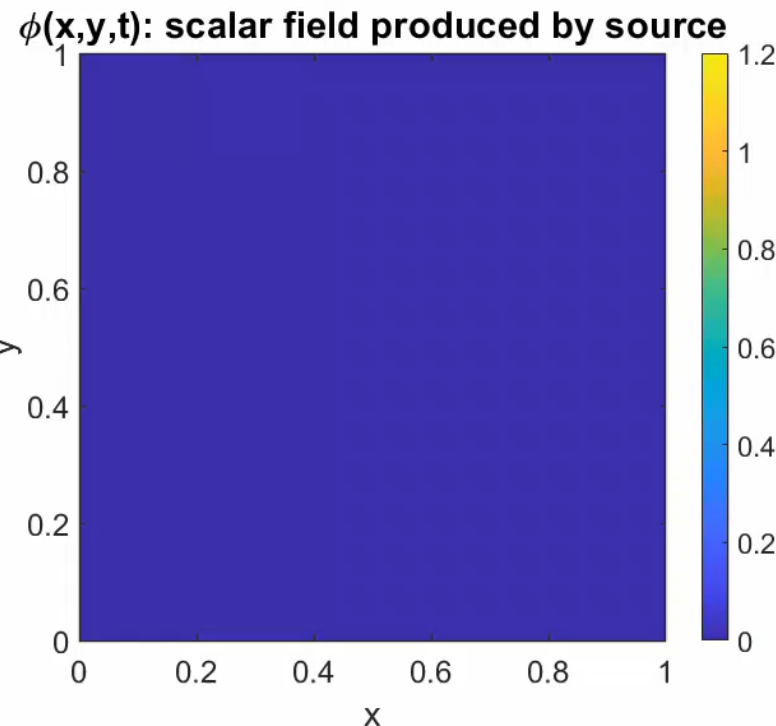
Types of Terms

Consider an equation of the form:

$$\frac{\partial(\phi)}{\partial t} + \frac{\partial(\phi u)}{\partial x} = \frac{\partial}{\partial x} \left(\Gamma \frac{\partial \phi}{\partial x} \right) + S_\phi$$

- Source Term (e.g., $S_\phi = \rho f_x$, $\frac{\partial p}{\partial x}$)
 - A local term defining creation or destruction of quantity ϕ at each point
 - For example, gravity creates momentum, a chemical reaction creates heat

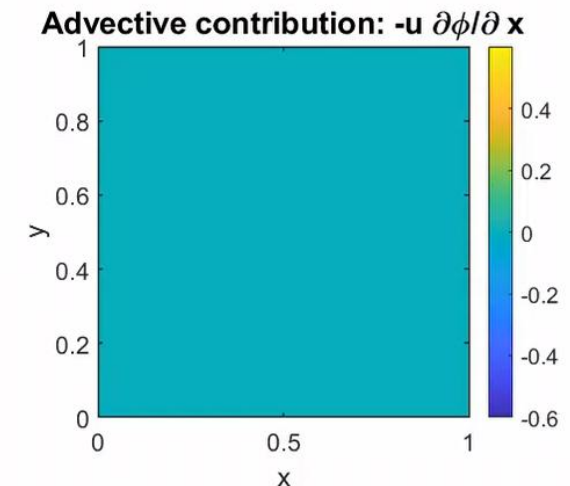
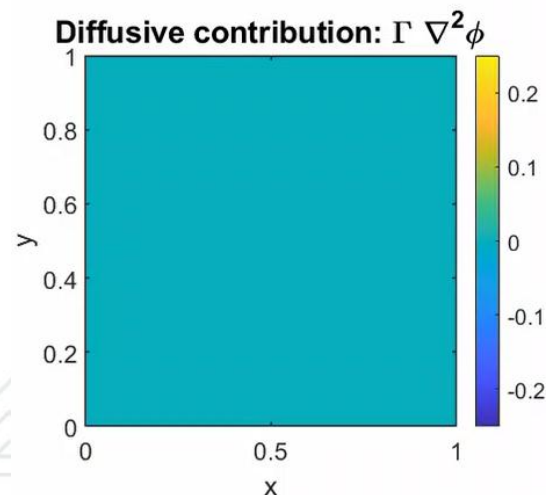
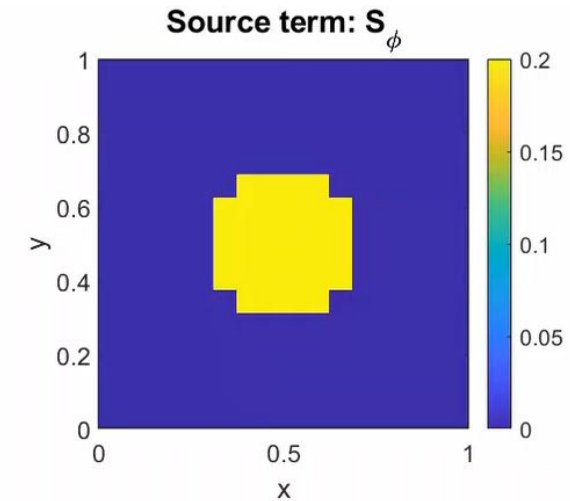
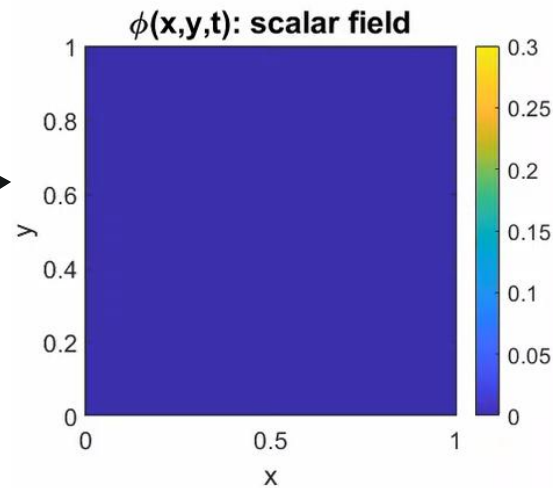
Unsteady term =
Source term



Types of Terms

- Bringing it all together

Unsteady term =
Advective term +
Diffusive term +
Source term



Looking Forward

- So far, we have derived four equations
 - One mass conservation equation
 - $f(\rho, u, v, w)$
 - Three momentum conservation equations
 - $f(\rho, u, v, w, P, \mu, f_x)$
- If we assume known forms for body forces and viscosity, we have five unknowns and four equations
 - Currently, the system is unsolvable and needs more equations
- Additionally, we have said nothing about energy
 - For example, when a nozzle accelerates a fluid (increases kinetic energy), where does that energy come from?
 - Must derive an energy balance